Efficient Synthesis of 2,3-Dihydrofurans via the Highly Selective Pd(0)-Catalyzed Coupling-Cyclization Reaction of 3,4-Allenols with Aryl Iodides

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Abstract: Highly selective coupling-cyclization of 3,4-allenols with aryl iodides via the oxidative addition-*exo*-mode oxypalladation-reductive elimination sequence afforded 2,3-dihydrofurans efficiently. The cyclization mode and the regioselectivity observed for 3,4-allenols are unique and different from what was reported previously.

Key words: alcohols, allenes, furans, palladium, cyclizations

Furans and dihydrofurans are not only widely present in natural products and non-natural compounds of pharmaceutical and agrochemical interest,[1,2] but are also versatile intermediates in organic synthesis.[3] Thus, much attention has been paid to the development of new methodologies for the synthesis of furans and dihydrofurans.[4] 2,3-Dihydrofurans can be prepared from a two-step procedure from γ-butyrolactone,[5] base promoted cycloisomerization of 3,4-allenols,[6] and conjugate addition of 3,4-epoxy-2-methylene oxolanes,[7] intramolecular enolate addition to 1-alkenyl sulfones,[8] base-mediated nucleophilic cyclization of 5,5-difluoro-4-enols,[9] addition reaction of 1,3-dicarbonyl compounds with alkenes,[10] transition metal-catalyzed coupling-cyclization of α-diazoketones with ethyl vinyl ether,[11] Pd-catalyzed cyclization of silyl-substituted bis(homo)propargylic alcohols,[12] Cr-mediated cycloisomerization of homopropargyl alcohols,[13] and the reaction of *p*-$TolSO_2Na$ with 1-hydoxybut-3-ynyl(phenyl)iodium triflate derivatives.[14] For all these methods, some of them require lengthy procedure, some need stoichiometric amount of metal complexes. Thus, the development of new and efficient methods for the synthesis of 2,3-dihydrofurans is still catching the attention of synthetic chemists. Recently, during the course of our studies of cyclization of functionalized allenes,[15,16] we have developed some new methodologies for the efficient synthesis of vinylic oxiranes,[17] and furans[18,19] starting from 2,3-allenols or 1,2-allenyl ketones. In this letter, we wish to communicate our recent observation on the Pd(0)-catalyzed coupling-cyclization reaction of 3,4-allenols with aryl iodides via the oxidative addition-*exo*-mode oxypalladation-reductive elimination sequence affording 2,3-dihydrofurans (Scheme).

During the course of our study of Pd(0)-catalyzed coupling-cyclization reaction of 2,3-allenols, we observed the exclusive formation of vinylic epoxides via either the carbopalladation of allenes followed by the intramolecular nucleophilic trapping of the in situ formed π-allyl palladium species ($n = 0$, pathway 1b, Scheme) or the Pd(II)-mediated *exo*-mode oxypalladation-reductive elimination (pathway 2, Scheme).[17] In principle, the possibility of forming 2,3-dihydrofurans ($n = 1$) exists, although we have not observed this pathway so far (pathway 3a, Scheme). Recently, when we switched from the chemistry of 2,3-allenols to that of 3,4-allenols, to our surprise, the exclusive formation of 2,3-dihydrofuran **2a** in 56% yield via $Pd(PPh_3)_4$-catalyzed coupling-cyclization reaction of 2-methyl-3-butyl-penta-3,4-dienol (**1a**) with PhI was observed (Equation and entry 3 of Table 1). It should be noted that the Lewis acid-catalyzed cyclization of 3,4-allenols afforded pyran derivatives.[20]

After careful examination of the crude reaction mixture, we found that the 6-membered product **3a** ($n = 1$, pathway

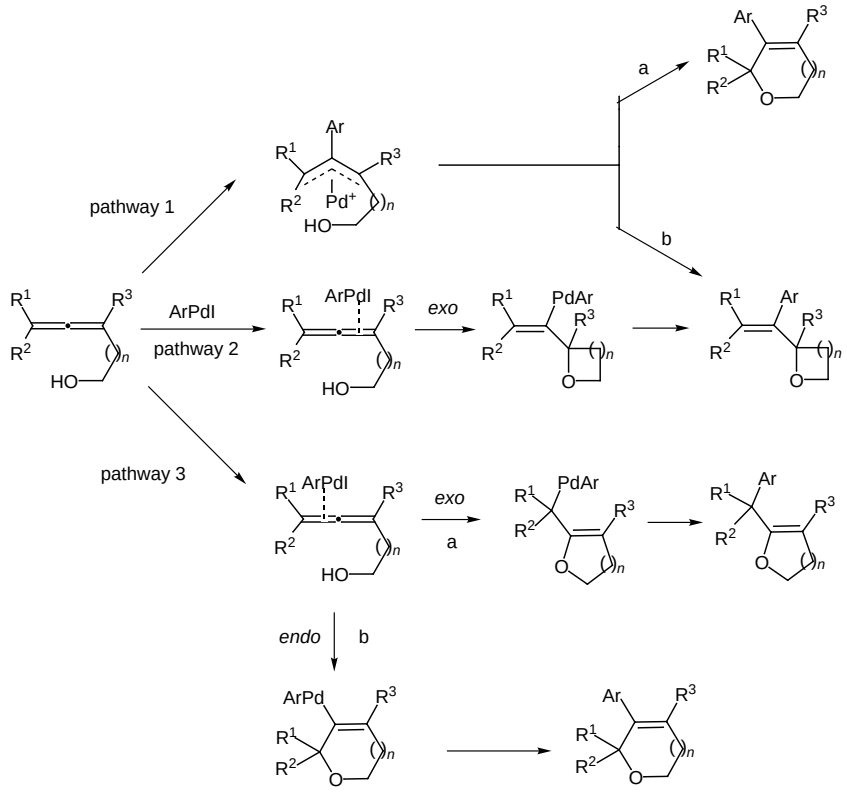
Table 1 Pd-Catalyzed Coupling-Cyclization Reaction of **1a** with Iodobenzene[a]

Entry	Base, 4 equiv.	Solvent	Temp. (°C)/(h)	Yield[b] (%)
1[c]	K_2CO_3	Toluene	80/11.5	Trace
2	Na_2CO_3	DMF	80/17	Trace
3	K_2CO_3	DMF	70/10	56
4[d]	K_2CO_3	DMF	60/11.5	42
5	Cs_2CO_3	DMF	70/4	67
6[c]	K_2CO_3	DMF	80/19	58
7[c]	K_2CO_3	THF	reflux/11.5	54
8	K_2CO_3	CH_3CN	80/17	26
9[c]	K_2CO_3	CH_3CN	80/17	54

[a] The reaction was carried out using **1a** (0.5 mmol), iodobenzene (1 mmol), base(2 mmol) and $Pd(PPh_3)_4$ (0.025 mmol, 5 mol%) in solvent (3 mL).
[b] Isolated yield based on **1a.**
[c] 1.4 equiv. of *n*-Bu_4NBr was added.
[d] 2.5 mol% $Pd_2(dba)_3{\cdot}CHCl_3$ and 10 mol% PPh_3 were used as the catalyst.

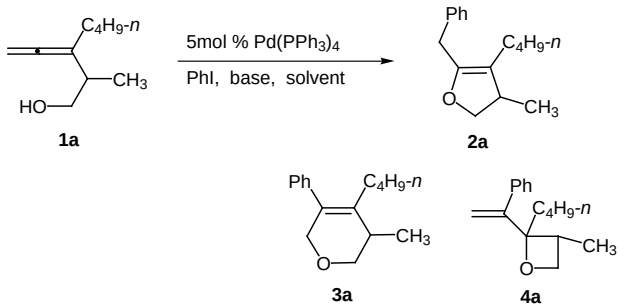
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Scheme

1a or pathway 3b, Scheme) and 4-membered product **4a** (*n* = 1, pathway 1b and pathway 2, Scheme) were not formed. From Table 1, the following points are noteworthy: (1) The reaction did not afford **2a** in decent yields in toluene using K_2CO_3 and *n*-Bu_4NBr as the base (entry 1, Table 1) or in DMF using Na_2CO_3 as the base (entry 2, Table 1); (2) The cation of the inorganic base has a very big impact on this reaction (compare entries 2, 3 and 5, Table 1); (3) The addition of *n*-Bu_4NBr has some influence for this reaction when the reaction was carried in CH_3CN using K_2CO_3 as the base (compare entry 8 with entry 9 of Table 1). Thus, we used 5 mol% $Pd(PPh_3)_4$, Cs_2CO_3 (4 equiv.) and DMF as the standard reaction conditions.



Equation

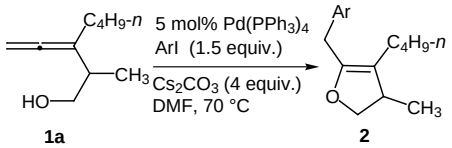
Some typical results of the Pd(0)-catalyzed reaction of 2-methyl-3-butyl-penta-3,4-dienol (**1a**) with differently substituted aryl halides are summarized in Table 2. The following points are noteworthy (1) For aryl halides with different carbon-halogen bonds, only the carbon-iodine bond was cleaved (entries 1–3, Table 2); (2) Both electron-donating and electron-withdrawing aryl iodides can be reacted to afford the corresponding products (entries 4–9, Table 2).

The scope of 3,4-allenols is summarized in Table 3. Based on the results in Table 3, it can be conduced that (1) a substituent at 3-position in 3,4-allenols is necessary since no expected cyclization products were obtained with 3-unsubstituted 3,4-allenols **1g** and **1h**.; (2) the substiuents at 1, 2 or 5-positions in 3,4-allenols are quite flexible.

In conclusion, we have developed an efficient synthesis of 2,3-dihydrofurans starting from the easily available 3,4-allenols.[21] Based on the structure of the products, it can be concluded that the reaction proceeds via a highly selective oxidative addition-*exo*-mode oxypalladation of the C-C double bond at 4-position-reductive elimination reaction, which is quite different from what we observed in our previous studies.[16–19] Further studies on this reaction including the factors controlling the reactivities of 3,4-allenols, the scope, and the synthetic application are being carried out in our laboratory.

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Table 2 Pd-Catalyzed Coupling-Cyclization Reaction of **1a** with Aryl Iodides[a]



1a **2**

Entry	ArI	Time (h)	2	Yield of 2[b] (%)
1	*p*-FC_6H_4I	4	**2b**	55
2	*p*-ClC_6H_4I	4	**2c**	68
3	*p*-BrC_6H_4I	4	**2d**	59
4	*p*-MeC_6H_4I	4	**2e**	68
5	*o*-MeC_6H_4I	6	**2f**	55
6[c]	*p*-$MeO_2CC_6H_4I$	10	**2g**	56
7	*p*-$MeO_2CC_6H_4I$	4	**2g**	60
8	*p*-$MeOC_6H_4I$	4	**2h**	63
9	naphthyl iodide	4	**2i**	58

[a] The reaction was carried out at 70 °C using **1a** (0.5 mmol), aryl iodide (0.75 mmol, 1.5 equiv), Cs_2CO_3 (2 mmol, 4 equiv) and $Pd(PPh_3)_4$ (0.025 mmol, 5 mol%) in DMF(3 mL).
[b] Isolated yield based on **1a**.
[c] K_2CO_3 was used instead of Cs_2CO_3.

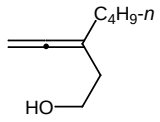
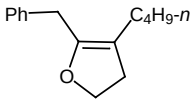
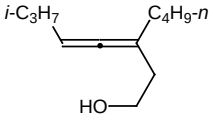
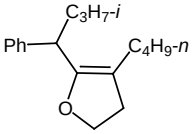
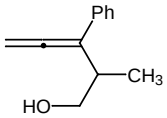
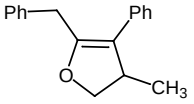
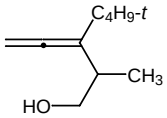
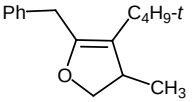
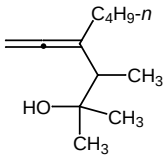
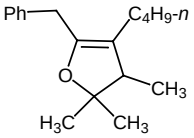
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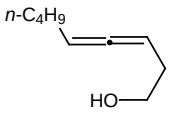
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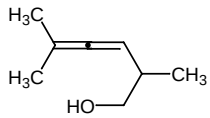
Table 3 Pd-Catalyzed Coupling-Cyclization Reaction of **1** with Iodobenzene[a]

Entry	Allenols **1**	Product **2** (%)[b]
1		
	1b	**2j** (75)
2[c]	**1b**	**2j** (58)
3		
	1c	**2k** (40)
4		
	1d	**2l** (61)
5		
	1e	**2m** (46)
6		
	1f	**2n** (43)
7[c]	**1f**	**2n** (50)

[a] The reaction was carried out at 70 °C using **1** (0.5 mmol), iodobenzene (0.75 mmol, 1.5 equiv), Cs_2CO_3 (2 mmol, 4 equiv) and $Pd(PPh_3)_4$ (0.025 mmol, 5 mol%) in DMF(3 mL).
[b] Isolated yield based on **1**. [c] KOH was used instead of Cs_2CO_3.



1g



1h

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